IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2023

MODELLING AND CONTROL OF MULTI-BODY MECHANICAL SYSTEMS

Friday 5 May 10:00 am

Time allowed: 3:00 hours

There are FOUR questions on this paper.

Answer ALL questions

All questions carry equal marks.

This is an OPEN BOOK examination.

Any special instructions for invigilators and information for candidates are on page 1.

Examiners responsible First Marker(s) : S.A. Evangelou
Second Marker(s) : D. Angeli

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MODELLING AND CONTROL OF MULTIBODY MECHANICAL SYSTEMS

1. A slider-crank mechanism consists of two bars, each of length l, mass m and negligible moment of inertia, and a slider of negligible mass, as shown in Figure 1.1. The first bar of the mechanism is fixed to Earth at point O and rotates in a vertical plane by an angle θ with respect to Earth about point O. The second bar is attached to the first bar at point A and to the slider at point B, and it rotates with respect to both the first bar (about point A) and the slider (about point B), also in a vertical plane. The slider is constrained to translate only along a horizontal line at the same height as O and its motion is frictionless. The slider-crank mechanism is driven by a motor torque M applied on the first bar. The centres of mass of the first and second bar are points C_1 and C_2, respectively.

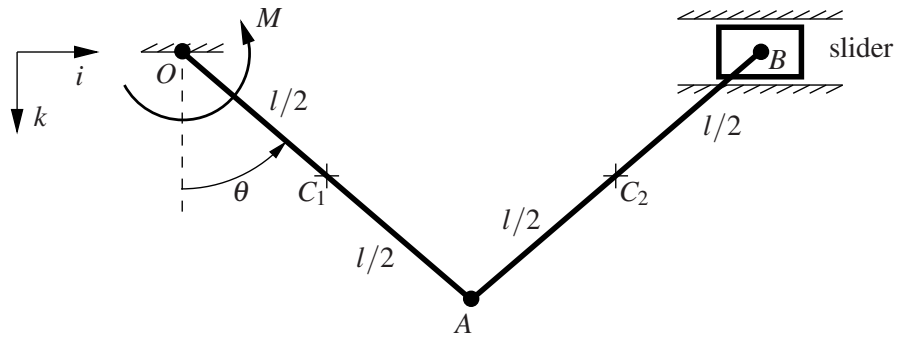


Figure 1.1 Slider-crank mechanism driven by a motor torque M.

A fixed Cartesian coordinate system with unit vectors $\boldsymbol{i}$ and $\boldsymbol{k}$ (see Figure 1.1) is used to analyse the motion of the slider-crank mechanism.

a) Write the number of degrees of freedom and the generalised coordinates of the system. [2]

b) Write

   i) the position vector, and [3]

   ii) the velocity vector [3]

   of the centre of mass of each bar, in the inertial reference frame.

c) Compute

   i) the kinetic energy, and [3]

   ii) the potential energy [3]

   of the system.

d) Derive the Lagrangian equations of motion of the slider-crank mechanism. [6]

2. A robotic cutting tool has three components, a vertical column (massless), a horizontal arm (massless and of length l) held by the column at point A, and a uniform axisymmetric disc (of mass m with spin moment of inertia I_{xx} and radial moment of inertia, passing through the centre of mass O, I_{zz}) held by the arm at point O, as shown in Figure 2.1. The column rotates relative to Earth about a vertical axis by an angle ψ, the arm rotates relative to the column about a horizontal axis by an angle θ, and the disc spins relative to the arm about a perpendicular axis to the arm by an angle ϕ. The spin axis of the disc is initially horizontal.

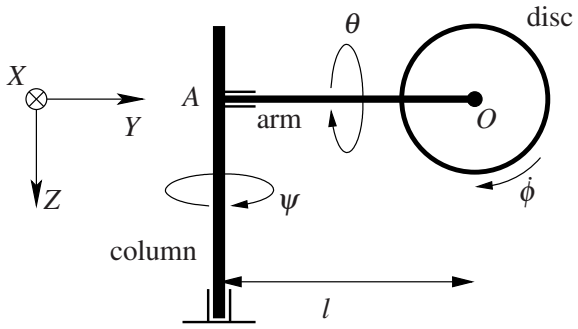


Figure 2.1 Robotic cutting tool.

An arm-fixed Cartesian coordinate system is used to analyse the motion of the system. This coordinate system has its origin at point O, it rotates with the arm and its unit vectors are initially, respectively, aligned with the earth-fixed axes X, Y and Z, in which X is into the page, as shown in Figure 2.1. The rotation of the arm is represented by two Euler angles ψ and θ in the yaw-pitch convention. The rotation of the disc is represented by three Euler angles ψ, θ and ϕ in the yaw-pitch-roll convention, in which ψ and θ are common with the arm rotation.

a) Write the number of degrees of freedom and the generalised coordinates of the system. [2]

b) By using the standard single-axis-rotation transformation matrices, determine the disc angular velocity vector in the arm-fixed coordinate system. [3]

c) Compute the kinetic energy of the system. [4]

d) Calculate the potential energy of the system. [2]

e) Derive the Lagrangian equations of motion of the system. [9]

3. As in Q2, a robotic cutting tool has three components, a vertical column (massless), a horizontal arm (massless and of length l) held by the column at point A, and a uniform axisymmetric disc (of mass m with spin moment of inertia I_{xx} and radial moment of inertia, passing through the centre of mass O, I_{zz}) held by the arm at point O, as shown in Figure 3.1. The column rotates relative to Earth about a vertical axis by an angle ψ, the arm rotates relative to the column about a horizontal axis by an angle θ, and the disc spins relative to the arm about a perpendicular axis to the arm by an angle ϕ. The spin axis of the disc is initially horizontal ($\theta = 0$).

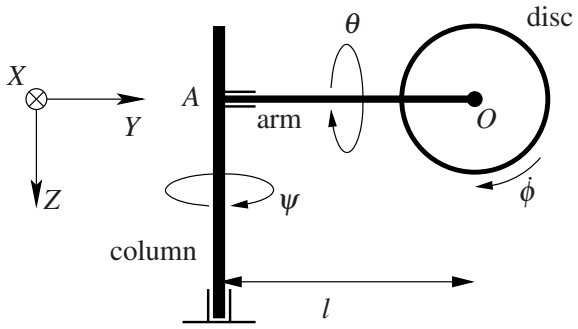


Figure 3.1 Robotic cutting tool.

An arm-fixed Cartesian coordinate system is used to analyse the motion of the system. This coordinate system has its origin at point O, it rotates with the arm and its unit vectors are initially, respectively, aligned with the earth-fixed axes X, Y and Z, in which X is into the page, as shown in Figure 3.1. The rotation of the arm is represented by two Euler angles ψ and θ in the yaw-pitch convention. The rotation of the disc is represented by three Euler angles ψ, θ and ϕ in the yaw-pitch-roll convention, in which ψ and θ are common with the arm rotation.

a) By using the standard single-axis-rotation transformation matrices, calculate the transformation matrices $\boldsymbol{A_1}$ from Earth-fixed to arm-fixed coordinates, $\boldsymbol{A_2}$ from disc-fixed to arm-fixed coordinates, and $\boldsymbol{A_3}$ from disc-fixed to Earth-fixed coordinates. [3]

b) Compute the angular momentum vector of the disc, $\boldsymbol{H}$, in the arm-fixed coordinate system. [4]

c) Use the vectorial approach in arm-fixed coordinates to derive the equations of motion of the system. [8]

d) Assume that initially the arm is static at $\dot{\psi} = 0$ and $\theta = 0$ (the disc spin axis is horizontal), and the disc spins with respect to the arm at a constant speed of $\dot{\phi} = \omega_{spin}$ about its spin axis. A moment M_y is then applied on the disc from the arm and tilts the spin axis of the disc from its initial angle $\theta = 0$ to an angle $\theta = \theta_t$ relative to the arm (relative to the horizontal). While the spin axis of the disc is maintained at $\theta = \theta_t$:

   i) calculate $M_y|_{\theta=\theta_t}$ (the moment M_y at $\theta = \theta_t$), [2]

   ii) determine the overall motion of the disc. [3]

4. As in Q1, a slider-crank mechanism consists of two bars, each of length l, mass m and negligible moment of inertia, and a slider of negligible mass, as shown in Figure 4.1. The first bar of the mechanism is fixed to Earth at point O and rotates in a vertical plane by an angle θ with respect to Earth about point O. The second bar is attached to the first bar at point A and to the slider at point B, and it rotates with respect to both the first bar (about point A) and the slider (about point B), also in a vertical plane. The slider is constrained to translate only along a horizontal line at the same height as O and its motion is frictionless. The slider-crank mechanism is driven by a motor torque M applied on the first bar. The centres of mass of the first and second bar are points C_1 and C_2, respectively.

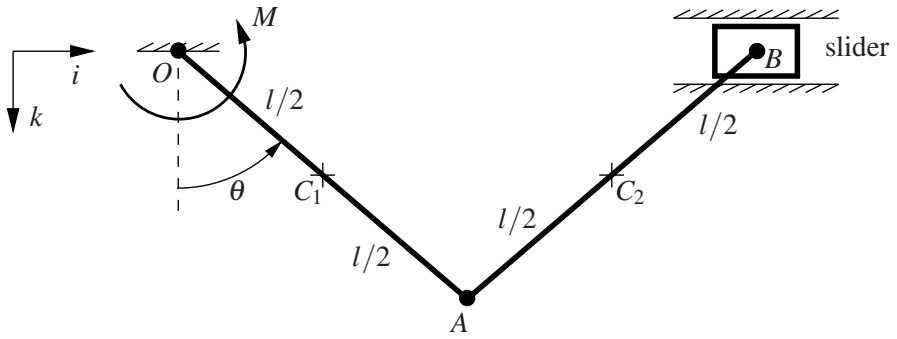


Figure 4.1 Slider-crank mechanism driven by a motor torque M.

A fixed Cartesian coordinate system with unit vectors $\boldsymbol{i}$ and $\boldsymbol{k}$ (see Figure 4.1) is used to analyse the motion of the slider-crank mechanism.

a) Write the acceleration vector of the centre of mass of each bar, in the inertial reference frame. [3]

b) Write the total force vector, in the inertial reference frame, that acts on each of the bars and includes any unknown constraint forces. [3]

c) By considering the translational motion of the centre of mass of each of the bars, use the vectorial approach to

   i) compute the horizontal component of the constraint force acting on the second bar at point A in terms of the generalised coordinates, [3]

   ii) compute the horizontal component of the constraint force acting on the first bar at point O in terms of the generalised coordinates. [3]

d) Hence, by considering also the rotational motion about the centre of mass of each of the bars, use the vectorial approach to derive the equation of motion of the slider-crank system in terms of the generalised coordinates. [8]